Cisco PE Node MG-PE-04 MG Road Core QoS Congestion Event Resolution SOP

Node: MG-PE-04 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_5064

Description

This Standard Operating Procedure (SOP) outlines the steps required to resolve a critical QoS congestion event on Cisco PE node MG-PE-04 located at the MG Road Core site. The failure domain is characterized by microbursts, interface tail drops, and QoS queue starvation, triggered by the ALARM_5064 alarm code. The Cisco PE node operates within a specific QoS framework, where the PE router's control plane and data plane are managed by the Cisco IOS XR software. The QoS configuration is based on the Modular QoS CLI (MQC) architecture, which enables the classification, policing, shaping, and queuing of network traffic. In this scenario, the QoS configuration has become misaligned, leading to congestion on the interface. The underlying cause of this misalignment is a result of a combination of factors, including incorrect QoS policy configuration, inadequate policing, or an excessive amount of traffic, which overwhelms the available bandwidth and causes the QoS queues to become starved.

Pre-requisite Checks

- pre-requisite: Verify that the MG Road Core site is accessible for remote hands and that the necessary credentials are available., status: Completed

- pre-requisite: Check the status of the redundant link to ensure it is operational., status: Completed

Potential Root Causes

Cause	Probability
Incorrect QoS policy configuration	High
Inadequate policing	Medium
Excessive traffic	Low

Diagnostic Commands & Expected Outputs

```
Command: show policy-map interface GigabitEthernet0/0/0/0, Expected Output: Policy Map is attached to the interface, Description: This command displays the QoS policy map attached to the GigabitEthernet0/0/0/0 interface. The expected output indicates that the policy map is attached, which may or may not be the root cause of the congestion event.

Command: show queue interface GigabitEthernet0/0/0/0, Expected Output: Queue is congested, Description: This command displays the queue statistics for the GigabitEthernet0/0/0/0 interface. The expected output indicates that the queue is congested, which supports the diagnosis of a QoS congestion event.
```

Resolution Steps

Step 1: Step: Verify the QoS policy configuration, Command: show policy-map, Description: Verify that the QoS policy configuration is correct and that it matches the expected behavior. If the configuration is incorrect, modify it according to the expected behavior.

Step 2: Step: Adjust policing, Command: policy-map type service POLICING, Description: Adjust policing on the interface to ensure that it matches the expected behavior. This may involve modifying the policer to allow for a higher or lower rate of traffic.

Step 3: Step: Modify QoS policy, Command: policy-map type service POLICY, Description: Modify the QoS policy to ensure that it matches the expected behavior. This may involve modifying the policy to ensure that it allows for a higher or lower rate of traffic.

Post-Check Verification

```
Command: show policy-map interface GigabitEthernet0/0/0/0, Expected Output: Policy Map is correctly attached to the interface, Description: This command verifies that the policy map is correctly attached to the interface and that it is functioning as expected.

Command: show queue interface GigabitEthernet0/0/0/0, Expected Output: Queue is no longer congested, Description: This command verifies that the queue is no longer congested and that the QoS congestion event has been resolved.
```

Roll-back Procedures

- Step: Restore previous QoS policy configuration, Command: copy running-config startup-config, Description: Restore the previous QoS policy configuration from the startup-config by copying it to the running-config.

- Step: Restore previous policing configuration, Command: policy-map type service POLICING, Description: Restore the previous policing configuration to its original state.

- Step: Restore previous QoS policy configuration, Command: policy-map type service POLICY, Description: Restore the previous QoS policy configuration to its original state.